Diagnosing Invalid Simplification of Sums of Radicals Answer Key

Grade 8 / Pre-Algebra — Irrational numbers and roots

Quick Answers

1. $6\sqrt{2}$
2. $6\sqrt{3}$
3. $5\sqrt{2}$, 7.07
4. 7, —
5. 13.19
6. $4\sqrt{3}$, 6.93
7. 4.24, —
8. —

Curriculum

Level: Grade 8 / Pre-Algebra — Irrational numbers and roots
8.NS.A – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–4 of 5 across 12 tagged checks.

Common wrong answers

3. *If they got 5.10: added the radicands and took one square root, writing sqrt(26)*
4. *If they got 5: added the radicands first and took a single square root, getting sqrt(25)*
5. *If they got 18.52: added the coefficients and the radicands together, writing 7*sqrt(7)*
6. *If they got 6: added and subtracted the radicands under one root, writing sqrt(36)*
7. *If they got 3.16: collected the two radicands under a single root, writing sqrt(10)*

1. Simplify $\sqrt{72}$.

Solution: Pull out the largest perfect-square factor. $72 = 36 \cdot 2$, and $\sqrt{36} = 6$:

$$\sqrt{72} = \sqrt{36 \cdot 2} = \sqrt{36}\,\sqrt{2}$$

$$\boxed{6\sqrt{2}}$$

The step that is legal here is splitting a root over a *product*. That is the rule students later over-apply to sums, which is what the rest of this sheet is about.

2. Simplify $5\sqrt{3} + 2\sqrt{3} - \sqrt{3}$.

Solution: All three terms carry the same radicand, so they are like terms and only the coefficients move: $5 + 2 - 1 = 6$. $\boxed{6\sqrt{3}}$

Treat $\sqrt{3}$ exactly as you would treat x: $5x + 2x - x = 6x$. The radicand never changes when like radicals are combined.

3. Simplify $\sqrt{8}+\sqrt{18}$.
Solution: Neither radicand is a perfect square, but both contain one. Simplify each term first:

$$\sqrt{8}=\sqrt{4\cdot 2}=2\sqrt{2}, \qquad \sqrt{18}=\sqrt{9\cdot 2}=3\sqrt{2}$$

Now the radicands match, so the terms combine: $2+3=5$. $\boxed{5\sqrt{2}\approx 7.07}$
Why not $\sqrt{26}$? Because $\sqrt{26}\approx 5.10$, and the sum is about 7.07. Adding the radicands is not a slower route to the same answer; it is a different number.

4. Find and fix: $\sqrt{9}+\sqrt{16}=\sqrt{25}$?
The mistake: the student used a rule that does not exist — $\sqrt{a}+\sqrt{b}=\sqrt{a+b}$. The real rule splits a root over a product, $\sqrt{ab}=\sqrt{a}\,\sqrt{b}$, and there is no matching rule for sums.
Correct work: take each root first, then add:

$$\sqrt{9}+\sqrt{16}=3+4$$

$\boxed{7>5}$

This is the cleanest counterexample in the whole topic, because both radicands are perfect squares and no decimals are needed: the true value is 7, the invented rule gives 5, and the gap is 2. A student who believes the rule can be asked to explain where the missing 2 went.

5. Find and fix: $4\sqrt{5}+3\sqrt{2}=7\sqrt{7}$?
The mistake: two additions happened at once. The coefficients were added $(4+3=7)$, which is legal only for like radicals, and the radicands were added $(5+2=7)$, which is never legal.
Correct work: check whether the radicands match after simplifying. Both 5 and 2 are already square-free and they are different, so the two terms are unlike and *nothing combines.* The expression is already in simplest form, and its decimal value is $4(2.2360\ldots)+3(1.4142\ldots)$. $\boxed{13.19}$
Watch for: 18.52, the decimal value of $7\sqrt{7}$. "Cannot be simplified" is a complete and correct answer here — students often assume an expression must reduce to a single term.

6. Simplify $\sqrt{12}+\sqrt{27}-\sqrt{3}$.
Solution: Simplify each radical before deciding anything:

$$\sqrt{12}=\sqrt{4\cdot 3}=2\sqrt{3}, \qquad \sqrt{27}=\sqrt{9\cdot 3}=3\sqrt{3}, \qquad \sqrt{3}=\sqrt{3}$$

All three share the radicand 3, so the coefficients combine: $2+3-1=4$. $\boxed{4\sqrt{3}\approx 6.93}$
Watch for: 6, which is $\sqrt{36}$ — what you get by adding and subtracting the radicands $12+27-3$. The near miss is instructive: 6 and 6.93 are close enough that a student who only estimates may not notice, which is why the simplified radical form is the answer of record.

7. Find and fix: $\sqrt{2}+\sqrt{8}=\sqrt{10}$?
The mistake: the radicands were collected under one root again. Here the two terms genuinely do combine — but through simplification, not addition of radicands.
Correct work: the decimal test first, since it settles the claim in one line: $1.4142\ldots + 2.8284\ldots$ gives $\boxed{4.24}$
while $\sqrt{10} \approx 3.16$. So

$$\boxed{\sqrt{2}+\sqrt{8} > \sqrt{10}}$$

For the exact form, $\sqrt{8} = 2\sqrt{2}$, so the sum is $\sqrt{2}+2\sqrt{2} = 3\sqrt{2}$, and $3\sqrt{2} = \sqrt{18}$ — not $\sqrt{10}$. The last line is the one worth showing: the sum of two radicals can be written as a single radical sometimes, and when it can, the number under the root is 18, not the sum 10 of the two radicands.

8. Explain: why roots do not add. *(Open response — flagged for manual review; a model answer follows.)*
Model answer: A square root asks which number, multiplied by itself, gives the radicand, and that question does not distribute over addition. Squaring $\sqrt{a}+\sqrt{b}$ gives $a+b+2\sqrt{ab}$, not $a+b$, so the two expressions differ by that middle term $2\sqrt{ab}$ — and it vanishes only when a or b is zero. That is the whole reason the invented rule fails, and it also shows the failure is not about "ugly" numbers: $\sqrt{9}+\sqrt{16} = 7$ while $\sqrt{25} = 5$, and every radicand there is a perfect square.
Two radical terms may be combined only when their radicands are identical after each has been simplified. Then they behave like like terms: $2\sqrt{2}+3\sqrt{2} = 5\sqrt{2}$, exactly as $2x+3x = 5x$. If the radicands still differ once simplified, as in $4\sqrt{5}+3\sqrt{2}$, the expression is already in simplest form and there is nothing to do.
What a reviewer should look for: identical radicands *after simplifying* named as the condition; a counterexample whose arithmetic is actually carried out; and no claim that the rule works for perfect squares and fails only for irrational roots — the perfect-square case is the counterexample.